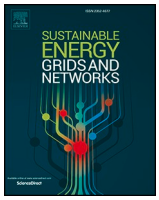
Sustainable Energy, Grids and Networks 43 (2025) 101889



Contents lists available at ScienceDirect

Sustainable Energy, Grids and Networks

journal homepage: www.elsevier.com/locate/segan



Optimization of pumped hydro storage management with renewable generation under wind uncertainty using chance-constrained programming

Natalia Naval *, Jose M. Yusta

University of Zaragoza, Department of Electrical Engineering, C/ Maria de Luna 3, Zaragoza 50018, Spain

ARTICLE INFO

Keywords:
Chance-constrained
Pumped hydro storage
Renewable energy
Management

ABSTRACT

The objective of this article is to develop an hourly mathematical model for the optimal techno-economic management of photovoltaic, wind and reversible pumped hydro storage plants in order to maximize the monthly operating profit over a whole year. The model incorporates the exchange of energy with the electrical grid to always meet the energy demand. In addition, the model includes the management of wind uncertainty through the chance-constrained technique, ensuring the coverage of the demand with renewable production in a certain percentage. The results obtained from the application of the model to a project under development in Spain show that the higher the risk of non-compliance with the chance-constrained technique constraints, the greater the flexibility of the system to optimally manage all available resources at minimum cost. In this way, although the system increases the imported energy up to 18 %, the costs are reduced up to 40 %, since the system leads to optimize the available resources, maximizing the profit. The system takes advantage of more competitive market prices compared to production prices at certain times. Therefore, maximizing the profit is prioritized over the coverage of demand with renewable resources.

1. Introduction

1.1. Context and generation and storage integration models

Renewable energies represent a challenge for the management of the electricity system due to their intermittency, which implies the need for greater storage capacity, improved cross-border interconnections and greater demand flexibility.

Among the storage technologies, pumped hydro power plants account for more than 90 % of the world's energy storage capacity. In Europe, there are 160 such plants with a total turbine capacity of 55 GW. The European Union countries have a total of 140 plants with a turbine capacity of 45 GW (installed pumped storage capacity is 39 GW), [1].

Research in the literature has explored the integration of reversible pumped hydro storage (PHS) with other sources of energy in order to provide a more stable and efficient energy supply. As indicated in the paper [2], this field has covered several directions such as technical, economic, techno-economic and energy management. Paper [3] reviews the technical and economic criteria as well as optimization methods for hybrid energy storage systems integrated with renewable energy. In addition, it proposes a comprehensive comparative framework that defines key performance indicators, aligns them with project goals or industry standards, normalizes the data, and calculates weighted scores to enable an objective and adaptable evaluation of energy storage systems.

Hybrid systems differ in the combination of energy sources they use, their coupling topologies and the integration of energy storage systems. In the reviewed literature, the most commonly studied configurations include PHS combined with photovoltaic energy [4], [5] or wind [6], [7] and in some cases, integrated with additional renewable or non-renewable energy sources [8], [9]. Work [4] studies the operation of PV-PHS systems, their characteristics, performance and feasibility. It also analyzes better configurations, operational strategies or designs to maximize efficiency, profitability and reliability. Paper [5] assesses the techno-economic feasibility of a hybrid PV system with grid-connected PHS in residential buildings. It studies performance by measuring indicators such as self-consumption rate and self-sufficiency rate. Article [6] proposes the optimal operation of a PHS-wind system. Its goal is to maximize profit by allowing cost-effective participation in the day-ahead market, while reducing variations in active power output due to intermittency of wind power. Work [7] focuses on optimal sizing of a PHS with wind generation in an island system. It adopts two perspectives: maximizing the return on investment, and maximizing the

* Corresponding author.
E-mail address: naval@unizar.es (N. Naval).

https://doi.org/10.1016/j.segan.2025.101889
Received 6 March 2025; Received in revised form 12 June 2025; Accepted 28 July 2025
Available online 30 July 2025
2352-4677/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Nomenclature

Indexes

h	index for number of hours
i	index for number of variable speed pump-turbines
y	index for number of time windows

Data

N_H	total number of hours of the calculation window
Nv	total number of variable speed pump-turbine machines
ρ^h_{exp}	hourly price of energy sales $\left(\frac{€}{MWh}\right)$
ρ^h_{imp}	hourly price of energy purchase $\left(\frac{€}{MWh}\right)$
f_{wind}	operating cost of wind power technology $\left(\frac{€}{MWh}\right)$
f_{PV}	operating cost of photovoltaic power technology $\left(\frac{€}{MWh}\right)$
$f_{pumping}$	operating cost of pumping $\left(\frac{€}{MWh}\right)$
$f_{turbining}$	operating cost of turbining $\left(\frac{€}{MWh}\right)$
$E^h_{PV,max}$	available photovoltaic generation (MWh)
$E^h_{wind,max}$	available wind generation (MWh)
E^h_d	hourly energy demand (MWh)
P^h_{cap}	grid connection capacity (MW)
$E^h_{exp_{max}}$	maximum hourly energy exported to the grid (MWh)
$E^h_{imp_{max}}$	maximum hourly energy imported from the grid (MWh)
$E^h_{storage,min}$	minimum energy of the storage system (MWh)
$E^h_{storage,max}$	maximum energy of the storage system (MWh)
$P^h_{turb,min,i}$	minimum power of variable speed units i in turbine mode (MW)
$P^h_{pump,min,i}$	minimum power of variable speed units i in pump mode (MW)
$P^h_{turbining,nom,i}$	rated power of variable speed machines i in turbine mode (MW)
$P^h_{pumping,nom,i}$	rated power of variable speed machines i in pump mode (MW)
$\eta^h_{pumping,i}$	hourly pumping efficiency of each variable speed unit i
$\eta^h_{turbining,i}$	hourly turbining efficiency of each variable speed unit i
k_{start}	pump-turbine start-up cost (€)

Variables

E^h_{exp}	hourly energy exported to the grid (MWh)
E^h_{imp}	hourly energy imported from the grid (MWh)
E^h_{PV}	hourly photovoltaic energy produced (MWh)
E^h_{wind}	hourly wind energy produced (MWh)
I^h_{exp}	binary variable equal to 1 if energy is exported to the grid, otherwise it will be equal to 0
I^h_{imp}	binary variable equal to 1 if energy is imported from the grid, otherwise it will be equal to 0
$I^h_{pumping,i}$	binary variable equal to 1 if a variable speed unit i is operating in pumping mode, otherwise it will be equal to 0
$I^h_{turbining,i}$	binary variable equal to 1 if a variable speed unit i is operating in turbining mode, otherwise it will be equal to 0
$I^h_{pumping}$	binary variable equal to 1 if the reversible machines are running in pumping mode, otherwise it will be equal to 0
$I^h_{turbining}$	binary variable equal to 1 if the reversible machines are running in turbining mode, otherwise it will be equal to 0
$I^h_{pumping,total}$	integer variable indicating the total number of machines in pumping mode in each hour
$I^h_{turbining,total}$	integer variable indicating the total number of machines in turbining mode in each hour
$I^h_{start_i}$	binary variable equal to 1 if the variable speed unit i is started, otherwise it will be equal to 0
$E^h_{storage}$	hourly energy stored in the system (MWh)
$E^h_{pumping,i}$	hourly energy pumped by each variable speed machine i (MWh)
$E^h_{turbining,i}$	hourly energy turbined by each variable speed machine i (MWh)
$C^h_{start,i}$	hourly start-up cost of variable speed pump-turbine i (€)
B	operating profit (€)
R^h_{exp}	hourly income from the sale of energy in the electricity market (€)
C^h_{imp}	hourly costs for importing energy from the grid (€)
$C^h_{pumping}$	total hourly pumping costs (€)
$C^h_{turbining}$	total hourly turbining costs (€)
C^h_{pv}	hourly photovoltaic production costs (€)
C^h_{wind}	hourly wind production costs (€)
C^h_{start}	total hourly machine start-up costs (€)

penetration of renewable energy as well as keeping the cost of generation as low as possible. The authors of [8] analyze the technical and economic feasibility of integrating solar, wind and pumped hydro storage to provide a reliable, clean energy supply in remote areas. The authors of [9] focus on optimizing the operation of a hybrid PV/Diesel/PHS system with the objective of minimizing diesel generator fuel consumption. This is achieved by optimally determining the share of the energy deficit that should be supplied by the PHS.

As pumped-solar-wind systems consist of different electrical power generation and storage components, the challenge is to obtain optimal system performance [10]. This is necessary to ensure that the maximum benefit is obtained and the costs of investment, operation and maintenance of the system can be recovered. To this end, it is essential to design an operational strategy that manages the complementarity of the system's components [11], [12], [13]. Optimal scheduling will depend on whether the system is isolated or connected to the electrical grid, whether it is open or closed- loop pumping, and other particular characteristics of each project. In [11], a pumped hydro storage system combined with offshore wind energy is proposed as a solution to improve energy management in offshore environments. Continuing along this line of research, work [12] analyzes the operational performance of a PHS system coupled with a grid-connected photovoltaic (PV) plant. Similarly, paper [13] examines the operation of a PHS system also connected to the grid, but integrated with wind energy.

Regarding participation in electricity market, article [14] proposes an iterative solution algorithm for the optimal sizing and long-term operation of a hybrid system (renewable generation and batteries) that participate in all segments of the wholesale electricity market (day-ahead market, reserve, and real-time balancing market). The authors of [15] develop an optimal two-stage energy management framework for integrated energy systems, with the participation in the day-ahead and real-time markets, addressing risk aversion and the handling of uncertainties in operating costs. Paper [16] focuses on the optimal scheduling of smart buildings with photovoltaic generation,

shared energy storage, and demand response. Its goal is to enhance economic viability by interacting with electricity markets for energy trading and managing tariffs to benefit from price volatility. The methods of solving the mathematical problems formulated to obtain the optimal scheduling of plant operation can be mainly classified into conventional methods, evolutionary algorithms and hybrid methods. Conventional methods include deterministic mathematical optimization, iterative, probabilistic, stochastic and analytical methods [17].

1.2. Uncertainty resolution methods

The accuracy of wind and solar power production forecasts has improved significantly in recent years, mainly for the solar resource, but wind power forecast errors currently range between 15 % and 20 % [18]. Stochastic optimization models thus become an alternative to consider the influence that uncertainty and forecasting errors of renewable energy generation have on the operating decisions of a production facility hybridized with an energy storage facility, overcoming the limitations of deterministic mathematical optimization.

The main feature of stochastic methods is that the uncertainty of the system input data is considered in the model, which increases the accuracy of the results [19], [20], [21], [22]. Work [19] uses historical data of wind speed. Article [20] uses a two-stage adaptive robust optimization approach to deal with wind uncertainty Paper [21] focuses on the management of wind uncertainty both in the planning of day-ahead market bids and in real time operation. The authors of [22] analyze wind and pumped hydro storage solutions for the normal operation of saturated insular power systems.

These works include wind farms along with pumped hydro storage and address the uncertainty of renewable production. A stochastic dynamic programming method for a reversible pumped hydro storage system that considers the uncertainty of the electricity price is presented in [23]. The results obtained are better than those of the deterministic method for the same model, which demonstrates the validity of the stochastic approach. In [24], a stochastic model for the optimal operation of a pumped-solar-wind system is developed, providing accurate storage and demand models. The results indicated that the stochastic model could improve system performance. However, stochastic techniques need a lot of data to ensure better distributed samples, which may require more computational time when dealing with a large number of input parameters [25], [26], [27]. In [25], [27], the authors deal with demand uncertainty in isolated systems using scenarios, while in [26], the authors use neural networks to predict the energy exchange in the hybridized system with the grid.

The most commonly used optimization methods with uncertainty are stochastic programming (SP), stochastic dynamic programming (SDP), robust optimization (RO) and probabilistic constraint programming (PCP). Optimization with uncertainty considers a set of variables or decisions to be made, a set of constraints that limit those decisions, and an objective function to be optimized. A solution is feasible in a deterministic optimization problem when it satisfies all the constraints, but when the constraints are random we cannot strictly speak of feasibility but rather of the probability that a certain solution is feasible. The objective function can be established as an expected value.

Stochastic programming (SP) can be used if the probability distribution of the random parameters is known. There are two-stage or multistage stochastic problems, and first-stage decisions are made before the uncertainty of the random parameters is addressed. In the next stages, scenarios are generated to capture these uncertainties, and decisions are made based on the probability of occurrence of the different scenarios. A stochastic optimization problem can be formulated by implicitly weighting (with the probabilities of occurrence) the individual solutions associated with each input data set to achieve a single solution that is best in some sense for all input data sets [28]. Stochastic problem solving can be computationally demanding, especially when there are a large number of scenarios. In [29], first-stage decisions are usually the selling prices of the energy produced, while operational decisions (power generation dispatch, energy storage, etc.) are usually second-stage decisions.

Stochastic dynamic programming (SDP) solves a problem by decomposing it into stages that are linked by recursive computations, so as to generate a feasible optimal solution for the entire problem. It has been commonly used in hydrothermal coordination problems, but also in problems of hybridization of renewables and pumped hydro storage [30], [31].

Robust optimization (RO) is an appropriate technique when the probability distribution of the random parameters is not known. Uncertainty is represented by data variation intervals and the problem is solved by deterministic optimization for worst-case scenarios. It requires less computational effort, but may provide less realistic solutions, since they correspond to unlikely situations related to the limits of the confidence interval of the data under uncertainty [32], [33]. Article [32] focuses on electricity market price uncertainty, while paper [33] addresses the uncertainties associated with renewable energy generation and load variability.

Stochastic programming with probabilistic constraints (CCP or chance-constrained) is a good tool for solving optimization problems with parameters subject to uncertainty. Some constraints may not always be satisfied due to some extreme events. Under chance-constrained optimization, these constraints with uncertainty can be satisfied with a certain probability. The objective function of an optimization model with probabilistic constraints is the same as that of the deterministic model. Thus, in the case of the optimal operation problem of a reversible pumped hydro storage, the main difference in the application of CCP is that the constraint set to limit the deviation of the net power produced in a pumped hydro storage is replaced by a probabilistic formulation in which can be satisfied with a given confidence level. On the other hand, if pumping is linked to the coverage of a certain electrical demand, it can be related to the number of hours in which it would be possible to deliver firm power with a certain probability. In works such as [34], [35] a CCP constraint is introduced in which the available wind energy is greater than the part consumed by the PHS installation and the part delivered to the grid with a certain probability. Paper [34] maximizes profits from the sale of energy in the electricity markets and minimizes penalties for operating imbalances. Article [35] proposes a stochastic short-term optimal dispatch model that includes detailed uncertainty modeling of multiple wind farms and PHS hydro constraints, aiming to maximize total generation profit. The authors of [36] use a CCP formulation in an intraday market optimization problem to consider the possibility of generating wind energy in a specific time interval and scenario without exceeding the expected wind power. In [37], the authors study the unit commitment problem considering probabilistic constraints for wind and photovoltaic energy demand and generation. Papers [38], [39], [18] use CCP to solve an economic dispatch problem with renewable energy and energy storage systems. The authors of [38] aim to ensure that a significant fraction of total demand is met by wind power with high probability. It highlights the reduction of computation time and its potential application in large-scale power systems. The authors of [39] focus on real-time dispatching and handling of distributional uncertainty without requiring precise distribution, using advanced linearization.

1.3. Gaps, objective, novel and contributions

The reviewed literature highlights a gap in the development of optimal hourly technical and economic dispatch models that integrate multiple renewable generation sources with long-term grid-connected pumped hydro storage. Although some studies address these aspects, they generally focus on short time periods and do not fully consider the uncertainty of renewable production. The high variability, especially in wind power, significantly affects the accuracy and reliability of long-term operational planning and scheduling. This limitation underscores

the need for more robust approaches to ensure efficient and stable energy system management over longer time horizons.

In this context, the problem that this article seeks to address is as follows: How does wind power uncertainty affect the operational decisions of hybrid generation facilities with energy storage? Therefore, this paper focuses on the proposal of an energy management system, that is, how operational decisions within the system are optimized to maximize the economic benefit of its operation.

The scientific objective of this work is to develop an optimal hourly scheduling model for closed-loop reversible pumped hydro storage plants to maximize the monthly operating profit in liberalized electricity market environments, with hybridization with wind and solar photovoltaic renewable generation. The model optimization is extended for a whole year. In addition, the uncertainty management of wind production is included, using the chance-constrained technique. The inherent variability of wind generation can significantly impact the profitability of these systems. Considering this uncertainty in planning allows adjusting operational strategies to maximize profit and ensure a more reliable supply of energy. Therefore, a robust model is achieved that allows for better decision making and optimization of the operation of pumped hydro storage power plants. In addition, data associated with a project under development in Spain for a pumped hydro storage integrated to wind and photovoltaic generation plants are used to assess the model's performance.

In summary, the main contributions of this research are as follows:

- Development of an optimal hourly scheduling model for closed-loop pumped hydro storage plants hybridized with wind and solar photovoltaic generation, aiming to maximize operating profit through dynamic energy trading strategies in the day-ahead electricity market.
- Incorporation of uncertainty in wind power generation using the chance-constrained technique. This approach improves the model's robustness by considering forecast errors, thus ensuring that technical constraints are met with a specified confidence level. Therefore, this technique enhances the reliability of the scheduling decisions.
- Assessment of the model in a real case over a full year, applying it to the optimal operation of a Spanish pumped hydro storage project, using realistic technical, economic, and electricity market data. This case demonstrates the model's practical applicability, its ability in order to handle operational constraints, and its potential in order to support investment and energy management decisions in the integration of large-scale storage and renewable generation systems.

Overall, this paper contributes to the advancement of energy management strategies in hybrid renewable-storage systems by providing a decision support tool that enhances economic performance while considering renewable intermittency and system constraints. Moreover, by applying the model to a project with real technical and economic data, it leads to demonstrate its practical feasibility and adaptability to real operating conditions.

The rest of the paper is structured as follows: Section 2 describes the outline of the proposed model and its approximations. Section 3 formulates the mathematical model with uncertainty. Section 4 defines the case study. Section 5 analyzes the results obtained. Finally, Section 6 summarizes the main conclusions of this work.

2. Description and limitations of the model

Fig. 1 shows the schematic of the model proposed in this article. The objective is to optimize the hourly management of the energy assets in a system that include a wind power plant, a photovoltaic plant and a reversible pumped hydro storage system. This management aims to meet the energy demand of industrial facilities, allowing energy exchange with the power grid in order to maximize the monthly operating profit.

The following assumptions have been considered for the development of this model:

- The model incorporates the possibility of purchasing and selling energy in the electricity market to meet demand at all times, and taking into account the connection capacity for exporting surplus production to the grid. The grid connection is established through an indexed pass-through electricity contract. A retailer operating in the Spanish electricity market offers this contract.
- Wind uncertainty using the chance-constrained technique to achieve a more robust model and evaluate its influence on the optimal operation of the system. This technique manages uncertainty by ensuring that certain model constraints are satisfied with a specified minimum probability. Instead of requiring that a condition must always be satisfied, it is accepted that it may occasionally be violated, but within a controlled level of risk, allowing for more flexible and realistic solutions in uncertain environments such as wind generation [37].
- Closed-loop pumped hydro storage is considered in this paper. Closed-loop pumped hydro storage uses a constant volume of water circulating between two reservoirs, without the need to draw additional water resources from the environment. By not depending on natural watercourses, this system minimizes environmental impact, preserves nearby ecosystems and offers a sustainable solution for renewable energy storage and management [40].

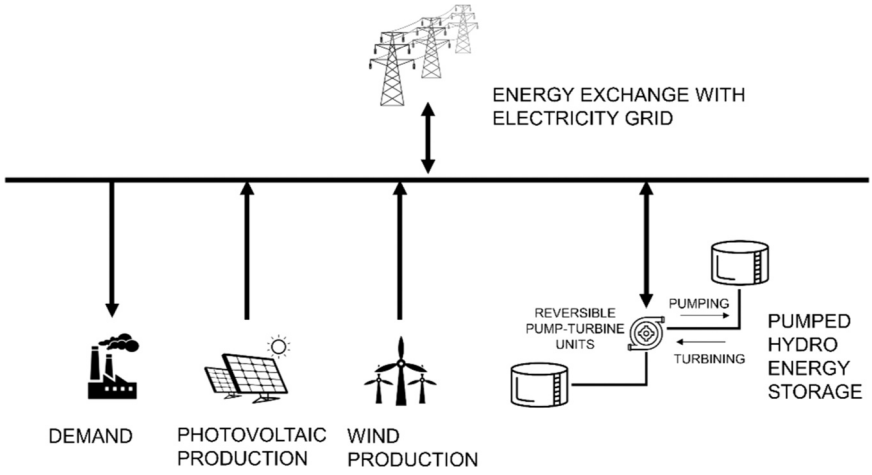


Fig. 1. Outline of the proposed model.

- The model only incorporates variable speed reversible pump-turbine machines. Compared to conventional fixed speed technology, variable speed drives provide faster response to load changes, greater efficiency and flexibility in pumping and generation modes, and improved operating and control characteristics [41].
- It is considered that reversible pump-turbine machines run in the same operating mode every hour to avoid the short-circuit of reversible machines with variable speed. Thus, it is avoided that some units operate in pump mode and others in turbine mode in the same hour.
- Pumping systems are commonly equipped with variable frequency drives, which allow the motor speed to be adjusted according to the needs of the system. Thanks to this technology, it is possible to operate the units at frequencies lower than the grid frequency (50 Hz), with 35 Hz being a commonly used minimum frequency. This value is usually applied to prevent pressure drops in the system and to maintain stable operation. The use of variable frequency drives not only provides operational flexibility, but also optimizes system performance, improves energy efficiency, and extends equipment lifespan by reducing mechanical wear and avoiding unnecessary overloads.
- The model assumes that evaporation losses do not affect system performance due to the proper design of the water storage and extraction system.
- The efficiency of the pump-turbine system is assumed constant over the entire operating range. According to the approach proposed by the authors of [42], performance variation can be disregarded when the operating speed varies by less than 33 % from the pump's rated speed, which supports the safe use of reduced frequencies such as 35 Hz without significantly affecting hydraulic performance.
- The problem considers the start-up costs of the pump-turbine units. These costs affect energy consumption, mechanical wear and tear and operating efficiency. Considering them is key to obtain the optimal technical and economic management of the system.
- A one-month optimization period is chosen to capture energy production patterns while maintaining a balance between planning accuracy and the model's computational feasibility. Additionally, one month is a reasonable period to achieve an adequate operational planning of available resources, which can lead to better decision-making.

3. Mathematical model formulation

The proposed hourly mathematical model aims to maximize the monthly operating profit for a year through the optimal management of the energy assets of the system, which include renewable generation plants and a reversible pumped hydro storage system. All components are connected to the electrical grid. The model considers the uncertainty management of wind generation, since compared to photovoltaic it has a higher degree of variability, and can affect more significantly the operation of the system [18].

3.1. Objective function

The objective function is the economic profit maximization function (B) resulting from the difference between income and costs of the system under study (Eq. (1)). Eqs. (2) - (8) identify each of the terms of the objective function. Index y indicates the calculation period. In this article, since the optimal operation of the system is considered for one month and the optimization is extended to a whole year, a period y is associated with one month.

Eq. (2) establishes the hourly income (R^h_{exp}) obtained from the sale of surplus energy generated to the grid ($\rho^h_{exp} \bullet E^h_{exp}$).

Eq. (3) defines the hourly cost of importing energy (C^h_{imp}) from the grid $\left(\rho^h_{imp} \bullet E^h_{imp}\right)$.

Eqs. (4) and (5) calculate the total hourly costs associated with pumping ($C^h_{pumping}$) and turbining ($C^h_{turbining}$) operations, respectively ($f_{pumping} \bullet \sum_{i=1}^{NV} E^h_{pumping,i}$/ $f_{turbining} \bullet \sum_{i=1}^{NV} E^h_{turbining,i}$).

Eq. (6) indicates the total start-up costs (C^h_{start}) of the reversible machines in each hour ($C^h_{start,total}$).

Finally, Eqs. (7) and (8) calculate the hourly PV (C^h_{pv}) and wind (C^h_{wind}) production costs, respectively ($f_{pv} \bullet E^h_{pv}$/$f_{wind} \bullet E^h_{wind}$).

$$\max((B)) = \sum_{y=1}^{Y} \sum_{h=1}^{N_H} \left(R^h_{exp} - C^h_{imp} - C^h_{pumping} - C^h_{turbining} - C^h_{start} - C^h_{pv} - C^h_{wind}\right)_y \quad (1)$$

$$R^h_{exp} = \rho^h_{exp} \bullet E^h_{exp} \quad (2)$$

$$C^h_{imp} = \rho^h_{imp} \bullet E^h_{imp} \quad (3)$$

$$C^h_{pumping} = f_{pumping} \bullet \sum_{i=1}^{NV} E^h_{pumping,i} \quad (4)$$

$$C^h_{turbining} = f_{turbining} \bullet \sum_{i=1}^{NV} E^h_{turbining,i} \quad (5)$$

$$C^h_{start} = C^h_{start,total} \quad (6)$$

$$C^h_{pv} = f_{pv} \bullet E^h_{pv} \quad (7)$$

$$C^h_{wind} = f_{wind} \bullet E^h_{wind} \quad (8)$$

3.2. Constraints

The sum of the hourly energy input and output in the system from the different sources must equal the energy demanded in each hour (E^h_d). As sources of energy input, we have imported energy from the electrical grid (E^h_{imp}), wind power generation (E^h_{wind}), photovoltaic (E^h_{pv}) y total energy turbined ($\sum_{i=1}^{NV} E^h_{turbining,i}$) from pumped hydro storage. On the other hand, as sources of energy output, the electrical energy exported to the power grid (E^h_{exp}) and total energy pumped to storage are considered ($\sum_{i=1}^{NV} E^h_{pumping,i}$), (see Eq. (9)).

$$E^h_d = E^h_{imp} - E^h_{exp} + E^h_{wind} + E^h_{pv} + \sum_{i=1}^{NV} E^h_{turbining,i} - \sum_{i=1}^{NV} E^h_{pumping,i} \quad (9)$$

Hourly renewable generation is limited by the maximum production availability of each technology, wind and photovoltaic ($E^h_{wind,max}$, $E^h_{pv,max}$). Thus, Eq. (10) limits the hourly wind power generation (E^h_{wind}) between 0 and the maximum production value, while Eq. (11) sets the hourly photovoltaic power generation (E^h_{pv}) limit between 0 and the maximum production value.

$$0 \le E^h_{wind} \le E^h_{wind,max} \quad (10)$$

$$0 \le E^h_{pv} \le E^h_{pv,max} \quad (11)$$

Eq. (12) ensures that electricity cannot be purchased and sold on the market at the same time within a given hour. To avoid this situation, two binary variables represent the decision making regarding the import

(I^h_{imp}) or export (I^h_{exp}) of energy in the electricity market. Therefore, there are three possible situations in each hour: energy is being imported ($I^h_{imp} = 1$, $I^h_{exp} = 0$), energy is being exported ($I^h_{imp} = 0$, $I^h_{exp} = 1$), or, there is neither export nor import ($I^h_{imp} = 0$, $I^h_{exp} = 0$).

$$I^h_{imp} + \quad I^h_{exp} \leq 1 \tag{12}$$

There are also hourly limits on the amount of energy imported and exported to the power grid. Eq. (13) states that, if the system is importing energy, and, therefore, the import binary variable, I^h_{imp}, takes value 1, the hourly energy imported (E^h_{imp}) cannot exceed the hourly energy demanded (E^h_d). On the other hand, Eq. (14) indicates that, if the system is exporting energy, the export binary variable, I^h_{exp}, takes value 1 and, then, the maximum energy that can be exported will be limited to the grid connection capacity ($P^h_{cap} \bullet \Delta t$).

$$0 \leq E^h_{imp} \leq I^h_{imp} \bullet E^h_d \tag{13}$$

$$0 \leq E^h_{exp} \leq I^h_{exp} \bullet P^h_{cap} \bullet \Delta t \tag{14}$$

Eq. (15) calculates the hourly storage level ($E^h_{storage}$) as the energy stored in the previous hour ($E^{h-1}_{storage}$) and the sum/subtraction of the energy pumped/turbined in the current hour, taking into account the efficiency of the machines in each of the operating modes.

$$E^h_{storage} = E^{h-1}_{storage} + \sum_{i=1}^{NV} E^h_{pumping,i} \cdot \eta^h_{pumping} - \sum_{i=1}^{NV} \frac{E^h_{turbining,i}}{\eta^h_{turbining}} \tag{15}$$

The storage level at the initial hour of the next calculation month ($E^h_{storage,y+1}(h = 1)$) must be equal to the storage level at the final hour of the previous month ($E^h_{storage,y}(h = N_H)$), (see Eq. (16)).

$$E^h_{storage,y}(h = N_H) = E^h_{storage,y+1}(h = 1) \tag{16}$$

It is considered that the storage level at the initial hour of the study period ($E^h_{storage,1} \quad (h = 1)$) is equal to the storage level at the final hour in the last period considered ($E^h_{storage,Y} \quad (h = N_H)$), (Eq. (17)). This constraint is considered to obtain a consistent operation scheme for the reservoir.

$$E^h_{storage,1} \quad (h = 1) = E^h_{storage,Y} \quad (h = N_H) \tag{17}$$

The storage system has a minimum and maximum capacity ($E^h_{storage,min}$, $E^h_{storage,max}$, respectively), so the energy stored in each hour must be between these two limits, (Eq. (18)).

$$E^h_{storage,min} \leq E^h_{storage} \leq E^h_{storage,max} \tag{18}$$

Regarding the constraints of the pumping/turbining units, a constant efficiency over their entire operating range is considered for both pumping $\eta^h_{pumping}$ and turbining $\eta^h_{turbining}$. The hourly turbined energy must lie between the minimum turbined energy ($P^h_{turbining,min,i} \bullet \Delta t$), as a lower limit, and the nominal energy of the machines in turbining mode ($P^h_{turbining,nom,i} \bullet \Delta t$), as an upper limit, also taking into account the efficiency of the machine in turbining mode (Eq. (19)):

$$\eta^h_{turbining} \bullet P^h_{turbining,min,i} \bullet I^h_{turbining,i} \bullet \Delta t \leq E^h_{turbining,i} \leq \eta^h_{turbining} \bullet P^h_{turbining,nom,i} \bullet I^h_{turbining,i} \bullet \Delta t \tag{19}$$

Eq. (20) shows the operating range of variable speed units in pumping mode. The binary variable ($I^h_{pumping,i}$) determines whether the unit is in pumping mode. $\eta^h_{pumping}$ is the hourly efficiency in pumping mode.

$$\left(\frac{P^h_{pumping,min,i}}{\eta^h_{pumping}} \right) \bullet I^h_{pumping,i} \bullet \Delta t \leq E^h_{pumping,i} \leq \left(\frac{P^h_{pumping,nom,i}}{\eta^h_{pumping}} \right) \bullet I^h_{pumping,i} \bullet \Delta t \tag{20}$$

The pumping ($I^h_{pumping,i}$) and turbining ($I^h_{turbining,i}$) mode cannot occur simultaneously in every hour, so the sum of the binary variables must always be less than or equal to 1, as can be seen in Eq. (21).

$$I^h_{pumping,i} + I^h_{turbining,i} \leq 1 \tag{21}$$

Eqs. (22)-(26) show the constraints for all the variable speed reversible machines in the pumped hydro storage system to run in the same operating state (pumping, turbining or shutdown), for this purpose, binary variables $I^h_{pumping}$, $I^h_{turbining}$ are defined. The parameter NV represents the number of variable speed units.

As mentioned above, these constraints are included to avoid short-circuiting of the variable speed reversible machines, i.e., that some of them operate in pumping mode and others in turbining mode in the same hour.

$$I^h_{pumping} + I^h_{turbining} \leq 1 \tag{22}$$

$$I^h_{pumping,total} = \sum_{i=1}^{NV} I^h_{pumping,i} \tag{23}$$

$$-NV \cdot \left(1 - I^h_{pumping}\right) \leq I^h_{pumping,total} \leq NV \cdot I^h_{pumping} \tag{24}$$

$$I^h_{turbining,total} = \sum_{i=1}^{NV} I^h_{turbining,i} \tag{25}$$

$$-NV \cdot \left(1 - I^h_{turbining}\right) \leq I^h_{turbining,total} \leq NV \cdot I^h_{turbining} \tag{26}$$

The binary variable $I^h_{start,i}$ is used to decide whether to start the pump-turbine units (Eq. (27)). The parameter k_{start} corresponds to the cost associated with machine start-up (Eq. (28)). Eq. (29) calculates the total start-up costs in each hour.

$$I^h_{pumping,i} - I^{h-1}_{pumping,i} + I^h_{turbining,i} - I^{h-1}_{turbining,i} \leq I^h_{start,i} \tag{27}$$

$$C^h_{start,i} = k_{start} \bullet I^h_{start,i} \tag{28}$$

$$C^h_{start,total} = \sum_{i=1}^{NV} C^h_{start,i} \tag{29}$$

To handle the wind power error, the chance-constrained method is used. This approach ensures that certain constraints are satisfied with a predefined probability level. Compared to the scenario generation method, the chance-constrained technique has several advantages: it enables better risk control and provides a more efficient approach by reducing the number of variables and, consequently, the calculation time. This is because it is not necessary to evaluate a large number of uncertainty scenarios to achieve reliable results [43]. As for solving the problem with chance-constrained constraints, it is appropriate to convert them into a deterministic formulation to reduce the complexity in solving the model.

The chance-constrained constraints of the proposed model are associated with demand coverage with renewables (Eqs. (30) and (31)):

$$\Pr\left(\sum_{h=1}^{Nh} E_{local}^{h} \le \gamma \cdot \sum_{h=1}^{Nh} E_{d}^{h}\right) \ge \alpha \tag{30}$$

$$\Pr\left(\sum_{h=1}^{Nh} E_{local}^{h} \ge \gamma \cdot \sum_{h=1}^{Nh} E_{d}^{h}\right) \ge \beta \tag{31}$$

In this model, the uncertainty of wind generation is considered, so these restrictions indicate the probability of demand coverage with local production E_{local}^{h} to be at least 1- α, where parameter α is the risk level when there is production deficit (Eq. (30)). Local generation, represented by E_{local}^{h}, corresponds to the renewable energy available in each hour within the system, considering only the part produced locally and that can be used directly or managed through storage. This variable is subject to uncertainty, especially in the case of wind generation. Therefore, parameter α is associated with the level of risk related to a shortfall in local generation. That is, the probability that the energy generated by renewable sources does not reach the minimum desired level to cover a proportion γ of the total demand. Hence, (1 - α) represents the minimum desired probability of meeting that demand coverage. A small value of α implies a conservative approach, where one seeks to ensure with high probability that renewable production covers at least the γ percentage of demand. On the other hand, Eq. (31) indicates the risk level β when there is excess production. Parameter β corresponds to the level of risk associated with an excess of local generation. In this case, (1 - β) indicates the maximum acceptable probability that renewable generation exceeds the coverage threshold γ of demand. This constraint is used to avoid producing too much more than necessary, which can generate losses or inefficiencies if there is insufficient storage capacity or possibilities of discharge to the grid. This provides some flexibility to manage uncertainty without always resorting to the grid or oversizing the storage system to meet the required demand. The parameter γ indicates the percentage of demand to be satisfied by the local renewable production of the system. The inclusion of this parameter allows achieving a balance between security of supply, operational efficiency and sustainability.

Next, the chance-constrained probabilistic constraints are transformed to deterministic constraints to facilitate the resolution of the problem.

It is assumed that the wind forecast error follows a normal distribution of zero mean and standard deviation proportional to the forecast value of wind generation $N(0, \sigma^2)$. A standard deviation equal to 10 % of the expected wind generation forecast has been used. According to [44], it is justified that this value covers approximately 99.73 % of the possible errors within the range ±30 % of the forecast, (Eq. (32)).

$$\varepsilon_{wind} = E_{wind,actual}^{h} - E_{wind,forecasted}^{h} \tag{32}$$

To convert Eqs. (33)-(35) into a deterministic constraint, the inverse of the cumulative density function (CDF) is used, as shown in Eq. (32). Furthermore, it should be noted that the wind prediction error is bounded between the limits a and b, corresponding to the lower and upper limit of wind generation capacity, respectively. Eqs. (34) and (35) correspond to the deterministic constraints.

$$F'(x) = \frac{F(x) - F(a)}{F(b) - F(a)} = \frac{F(x) - F(-E_{wind,forecasted}^{h})}{F\left(E_{wind,max}^{h} - E_{wind,forecasted}^{h}\right) - F(-E_{wind,forecasted}^{h})} \tag{33}$$

$$\sum_{h=1}^{Nh} E_{local}^{h} \le \gamma \cdot \sum_{h=1}^{Nh} E_{d}^{h} + F'^{-1}(\alpha) \tag{34}$$

$$\sum_{h=1}^{Nh} E_{local}^{h} \ge \gamma \cdot \sum_{h=1}^{Nh} E_{d}^{h} - F'^{-1}(\beta) \tag{35}$$

$F'^{-1}(\alpha)$ corresponds to the inverse of the conditional CDF of the prediction error when there is a generation deficit, calculated for a confidence level α. $F'^{-1}(\beta)$ is the inverse of the conditional CDF of the prediction error when there is excess generation, calculated for a confidence level β.

From the formulation of the mixed-integer mathematical model, GAMS (General Algebraic Modeling System) has been used for the optimization of this problem, using the SCIP solver. The SCIP method (Solving Constraint Integer Programs) is a solver specialized in solving mathematical programming models involving integer, mixed and constrained variables. SCIP's central approach is based on the branch and bound method, which is an exploratory search technique used to find the optimal solution to an optimization problem. Through branching, the original problem is divided into smaller subproblems, while bounding eliminates those that cannot produce a solution better than the best one currently identified. SCIP has high performance in solving large and complex optimization problems.

4. Case study

The case study to analyze the behavior of the proposed mathematical model consists of a Spanish project composed of wind, photovoltaic and reversible pumped hydro storage plants, all connected to the grid connection to meet the electricity demand of an electro-intensive industry. Spain's varied orography, with its mountainous terrain and extensive hydrographic network, provides ideal conditions for this type of storage. Driven by the increasing adoption of renewable energy and the need for more efficient electricity management, numerous pumped hydro storage projects are currently being developed across various regions of the country. These initiatives will help contribute to a more flexible, reliable, and environmentally sustainable energy system. Table 1 indicates the data of the case study. A Spanish company, dedicated to the development of wind and photovoltaic plants, has provided the technical and economic input data. In this case, the demand satisfaction of an electro-intensive industry has been considered, so a fixed demand has been used to assess the results from the proposed model.

Figs. 2 and 3 represent the monthly photovoltaic and wind power production of the system's power plants, respectively.

Table 1
Data of case study.

Electricity demand	396 MW
Grid connection capacity	396 MW
Rated wind power	465 MW
Rated photovoltaic power	860 MW
Units	4 variable speed pump-turbine units
Minimum turbine power	33.96 MW
Rated turbining power	99 MW
Minimum pumping power	38.93 MW
Rated pumping power	113.5 MW
Pump-turbine system efficiency (hydraulic efficiency, motor and transformer efficiency)	0.9
Pumped hydro storage capacity	5750 MWh
Pump-turbine start-up cost	10 € [45]
Wind power operation and maintenance cost	15 €/MWh
Photovoltaic power operation and maintenance cost	8 €/MWh
Pumping/Turbining operation and maintenance cost	3.5 €/MWh
Demand coverage with local production	60 %
Power purchase/sale prices in the Spanish electricity market OMIE	Year 2023/ The purchase of energy is established through a contract indexed to electricity prices in the wholesale market. Sales prices are associated with hourly prices in the wholesale electricity market, considering an electricity generation tax of 7 % [46]

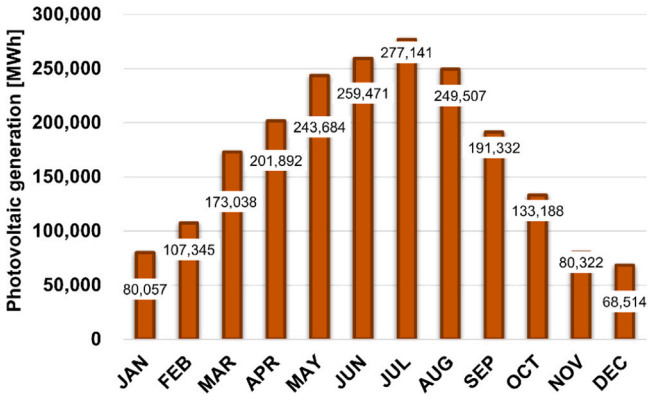


Fig. 2. Monthly photovoltaic generation.

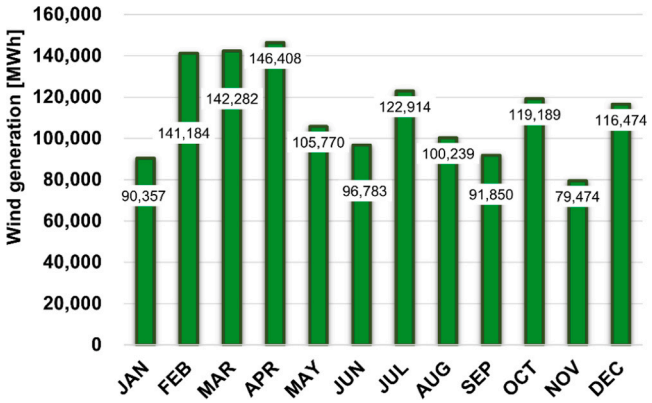


Fig. 3. Monthly wind generation.

5. Results and discussion

The model proposed in this paper aims to obtain the optimal hourly management of renewable resources, storage and exchange with the grid to meet a demand while maximizing the monthly operating profit. In addition, the model incorporates wind uncertainty management, using the chance-constrained technique. It is considered to meet the demand coverage with renewable generation at a certain percentage depending on the level of risk allowed in the model.

The optimization model is a problem with xxx continuous variables and xxx integers in each hour. The optimization model calculates the optimal hourly values of the binary integer variables for decision making in the problem:

- Import/Export of energy I^h_{imp}/ $I^h_{\exp}$
- Operation mode of each machine: Pumping/Turbining $I^h_{pumping,i}$/ $I^h_{turbining,i}$
- Operation mode in each hour in order to avoid the short-circuit of these machines: Pumping/Turbining $I^h_{pumping}$/$I^h_{turbining}$
- Start-up of the pump-turbine units $I^h_{start_i}$

Furthermore, the model calculates the optimal hourly values of:

- Electricity production from renewable energy plants: wind and photovoltaic E^h_{wind}, E^h_{PV}
- Energy imported/exported to the grid E^h_{imp}, $E^h_{\exp}$
- Level of pumped hydro storage $E^h_{storage}$
- Energy pumped/turbined of each reversible machine $E^h_{pumping,i}$, $E^h_{turbining,i}$

The chance-constrained method is a technique for handling uncertainty in optimization problems. In this approach, some constraints are allowed to be unfulfilled under certain circumstances, but with a controlled and limited probability. This allows managing uncertainty in real systems, where it is difficult to guarantee that the constraints are always met due to stochastic variations in the data, as in the case of renewable energy [47].

This technique has several advantages such as its flexibility, since it establishes a desired probability that a constraint will be satisfied, leading to adjust the confidence level according to the specific needs of the problem. In addition, the complexity of the model can be reduced in terms of number of variables and constraints with respect to scenario generation which requires a large number of scenarios to adequately manage uncertainty and, therefore, this increases the computational complexity, [29], [48]. In addition, it highlights that once the model has been formulated, with the chance constrained method it can be treated as a deterministic problem with tight constraints, which allows simplifying the optimization process and obtaining a solution by means of standard mathematical programming techniques, as used in works [49], [44].

The authors of [44] compare both methods for solving the daily operation optimization problem of a PHS-Wind system and greater benefits are obtained with the chance-constrained method since here it only requires that Wind-PHS meets the power deviation limit with a given probability, which has a larger feasible region, to obtain a better result. Similarly, the authors of [48] conclude that the chance-constrained method improves computational efficiency and scalability significantly, in addition to obtaining better utilization of renewable energy and a more reliable constraint on operational risks.

Different confidence thresholds are considered, as shown in Table 2.

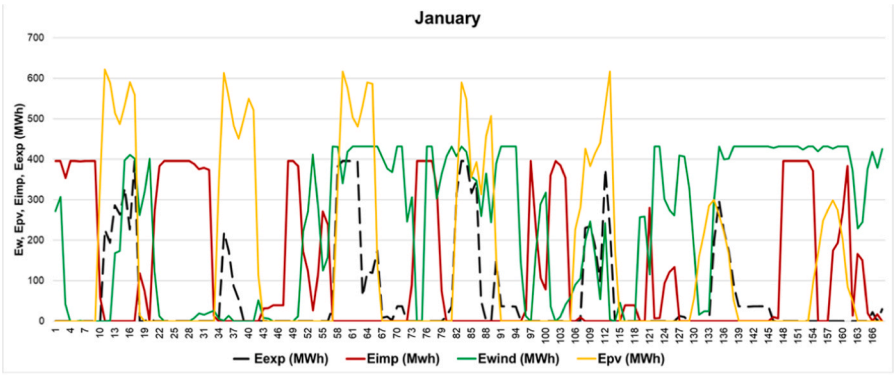
5.1. Hourly energy management results

First, the hourly results of energy management are analyzed for the proposed Case 1 (see Fig. 4). Hourly data for one week of different months of the year are shown to evaluate the performance of the dispatch model (January, Fig. 4a, April, Fig. 4b, July, Fig. 4c, October, Fig. 4d). A week has been represented instead of a full month obtained from the optimization to better visualize the results obtained. Furthermore, Fig. 5 shows the energy purchase and sale price profiles for the corresponding hourly results analyzed for each week (January, Fig. 5a, April, Fig. 5b, July, Fig. 5c, October, Fig. 5d). For the hourly analysis of the model performance, a week in January, April, July and October is considered to reflect the seasonal variability in renewable availability. These months are associated with different seasons and renewable source profiles, hence providing a better assessment of the optimal energy management of the proposed hybrid system.

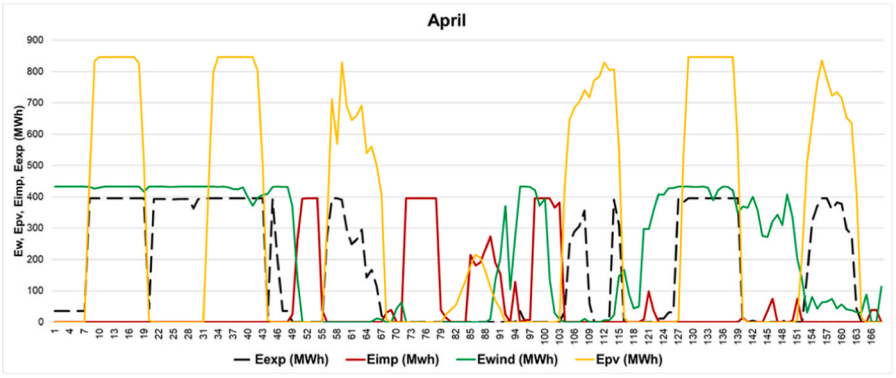
Photovoltaic energy is produced during daylight hours, so its availability and the amount generated changes according to the season of the year. However, wind energy has a great variability, so it generates a large amount of surplus energy. As a general trend, photovoltaic generation shows a seasonal increase, peaking in the months of highest solar irradiation (mainly July, exceeding 800 MWh on several days analyzed).

Table 2
Proposed study scenarios.

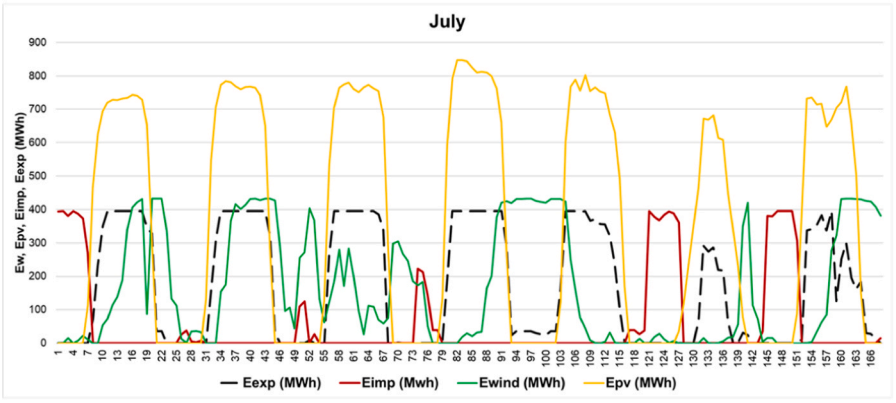
Case	100-α (%)	100-β (%)
1	95 %	90 %
2	90 %	85 %
3	85 %	80 %
4	80 %	75 %
5	75 %	70 %



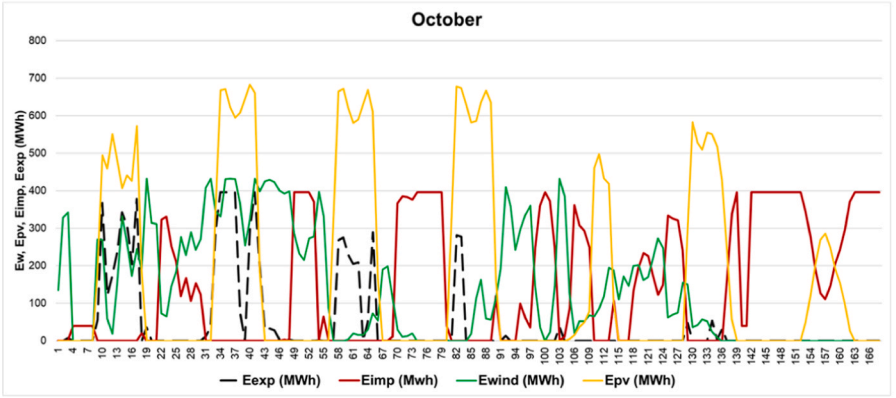
a) Generation, export and import electricity for one week in January



b) Generation, export and import electricity for one week in April

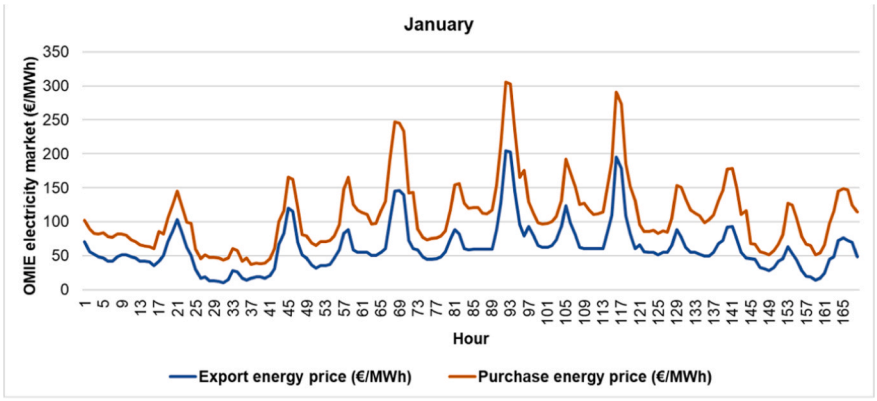


c) Generation, export and import electricity for one week in July

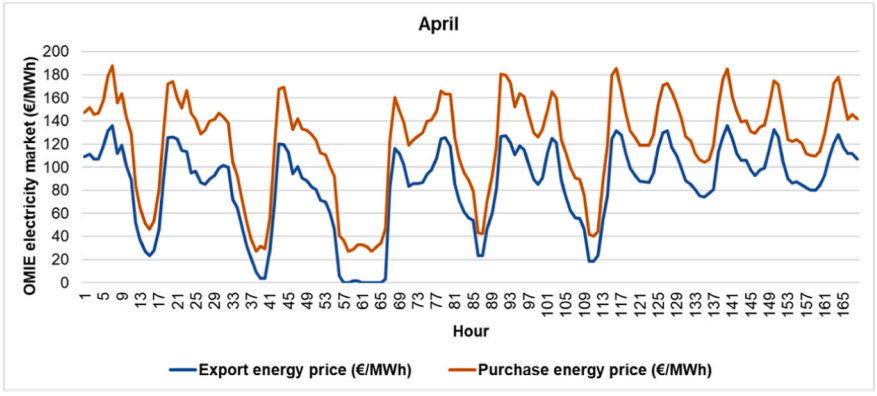


d) Generation, export and import electricity for one week in October

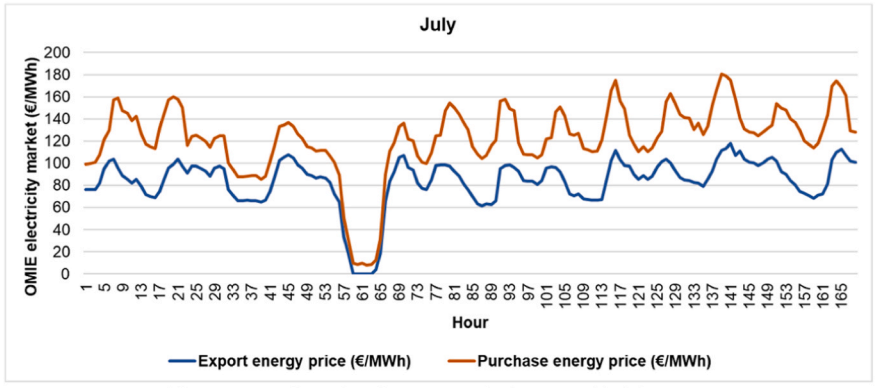
Fig. 4. Optimal hourly energy results for one week of different months (Case 1).



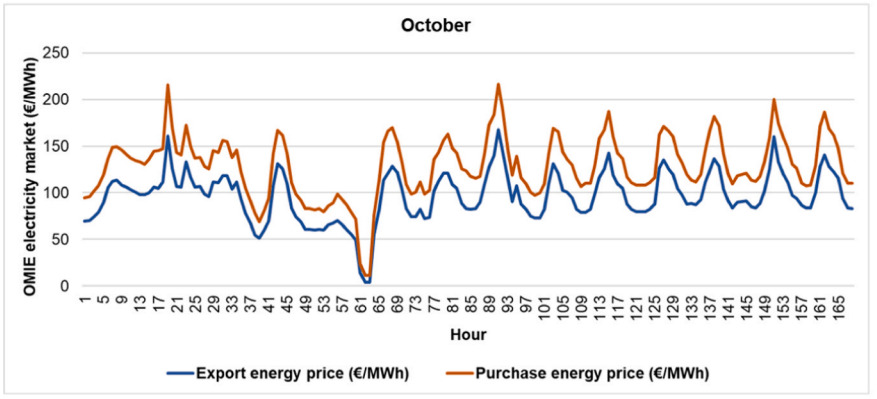
a) Export energy price and purchase energy price for one week in January



b) Export energy price and purchase energy price for one week in April



c) Export energy price and purchase energy price for one week in July



d) Export energy price and purchase energy price for one week in October

Fig. 5. Export/purchase energy prices for a week of different months of 2023 (Case 1).

In contrast, in January, the values are significantly lower due to the lower availability of sunlight, reaching a maximum of approximately 622 MWh. Similarly, this decrease is seen due to seasonal change, which marks the onset of solar decline towards winter. In addition, wind generation presents greater variability, although with recurring maximum values around 432 MWh. In January, daily peaks occur intermittently, while in April and July, wind generation reaches similar values, but with a more stable distribution throughout the month. In months with peak solar generation like April and July, wind energy production remains steady but plays a reduced role as photovoltaic output becomes dominant. An important trend is the complementarity between both renewable sources: when photovoltaic generation is low, wind generation tends to cover part of the demand, and vice versa. This combination contributes to reducing the need for purchasing energy in the electricity market and increasing export capacity.

Due to the intermittency and variability of renewable generation, together with fluctuations in the electricity price, a variation of imported energy is produced depending on the months of the year. It should be mentioned that the proposed model should always satisfy the demand. Energy imports (E_{imp}) are notable during January, with daily values of up to 396 MWh, especially during the first days of the month to fully meet electricity demand. This reflects a greater dependence on external sources due to lower renewable generation. As can be seen, since July corresponds to a month where photovoltaic generation is high, imports are minimal, with days without the need for purchasing energy in the electricity market to satisfy the demand. Similarly, in April, where local generation is higher and can meet demand without the need to import from the grid. Furthermore, a large amount of surpluses are produced and used to fill the storage, or to sell these surpluses directly on the electricity market to obtain additional income. Energy exports (E_{exp}) follow an increasing trend. In January, the maximum daily values reached 396 MWh, while in April and July, this value remained constant, indicating a stable export capacity. October shows a decrease in export levels. Generally, without reaching the maximum allowed export levels. The greater availability of renewable energy makes it possible to generate surpluses that can be sold to the day-ahead electricity market, contributing to the profitability of the energy system.

In short, the system shows a good performance in terms of renewable resource utilization, especially during spring and summer. However, January represents the weakest point, where the low solar power supply makes it necessary to depend heavily on imports. Therefore, the usefulness of the proposed model of optimal technical-economic dispatch can be verified, which allows an adequate coordination of all available resources of the system to maximize the profit and guarantee a sustainable and efficient energy management.

On the other hand, Fig. 6 represents the optimal pumped hydro storage management during one week for different months of the year (January, Fig. 6a, April, Fig. 6b, July, Fig. 6c, October, Fig. 6d). In the same way as for energy results, a week has been represented instead of a full month obtained from the optimization to better visualize the results obtained.

Storage plays a key role in managing the variability of renewable sources, making it possible to take advantage of surplus generation and guarantee a stable supply during periods of lower production.

Pumping shows an increasing trend from January to July, reflecting a greater use of the surpluses generated, especially during the months with higher photovoltaic production. Average daily pumping increases from 193 MWh/day in January to 326 MWh/day in July. In April, pumping reaches high values for more consecutive days, taking advantage of the higher surplus of photovoltaic generation. The stability in peak values reflects a greater capacity to store energy on a continuous basis.

Regarding the turbining mode, the analyzed values present a progressive growth for energy turbined, reflecting greater utilization of stored energy to maintain a stable supply. The daily average increases gradually from 176 MWh/day in January to 291 MWh/day in July, indicating greater use of stored energy as renewable generation increases and export capacity rises.

A relevant aspect is the use of storage to balance the variability of wind and PV generation. For example, during days with high PV generation, machines operate in pump mode to store surplus energy, while on days with lower generation, reversible units run in turbine mode to cover demand. This strategy helps to reduce dependence on imports and maximize system efficiency.

In addition, variable pump-turbine machines allow for precise adjustment of pumping and generation times based on fluctuations in the electricity market prices (See Fig. 5). Their ability to operate at partial load and dynamically adapt to system conditions enables the maximization of economic benefit: water is pumped during low-price periods, which generally coincide with hours of high renewable generation, especially from photovoltaic sources, and energy is generated during high-price hours, taking full advantage of tariff differentials. Thus, these machines not only ensure technically efficient operation of the storage system but also contribute to the economically optimal performance of the integrated energy system.

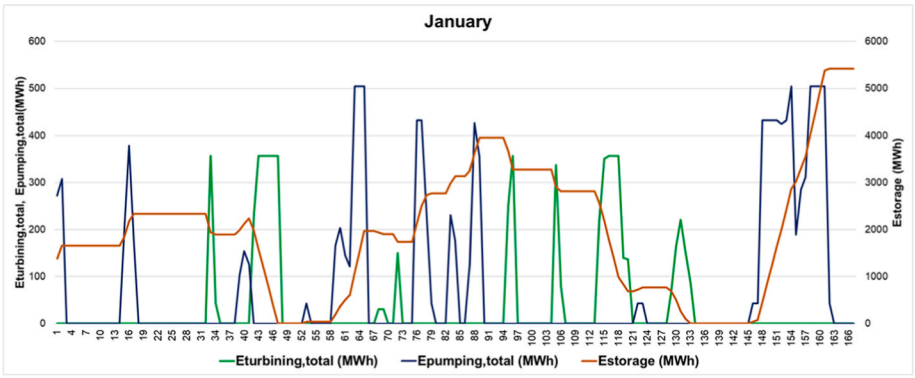
5.2. Annual energy results

To illustrate the monthly energy behavior of the system, Fig. 7 represents the monthly energy results for Case 1 of the study. As can be seen, the system operation is largely influenced by the seasonality of renewable sources, especially photovoltaic energy in summer, and wind energy in spring and autumn. The months of June and July, with higher photovoltaic generation (250,119 MWh and 245,195 MWh, respectively), resulted in a lower dependence on the grid to meet the required demand (imports were reduced by 76 % with respect to the highest value obtained in January). Consequently, the use of storage (a 50 % boost in pumping/turbining compared to the minimum value obtained in the winter months) and exported energy (a 90 % increase) is raised to take advantage of electricity market prices, the large amount of surplus photovoltaic and wind energy, and so, maximize the system's operating profit.

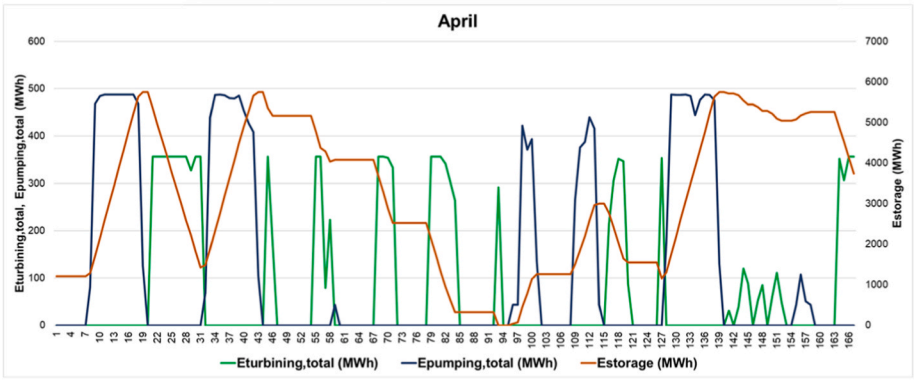
Table 3 presents the optimal annual energy results obtained for the different cases proposed. As can be seen, by increasing the allowed risk, i.e., accepting a higher probability of not meeting 60 % of the demand with renewables, photovoltaic and wind generation decreases. The system becomes more flexible, since the model chooses to import a greater amount of energy from the grid at times when the market price is more competitive than the production prices of renewable power plants, thus maximizing the operating profit of the system. Imported energy increases by 18 % from Case 1 (1,158,736 MWh) to Case 5 (1,413,979 MWh), which contributes to maximizing operating profits. In parallel, exported energy decreases by 42.3 % (from 473,731 MWh to 273,203 MWh), reflecting the reduced need to export surpluses.

Regarding the use of pumped hydro storage, as the allowable risk (α, β) increases, its use decreases. In general, pumped hydro storage is used to shift surplus renewable energy to more profitable times, or to ensure that demand is met at critical times. Total energy pumped decreases by 27.8 % between the extreme cases, falling from 810,271 MWh in Case 1 to 584,718 MWh in Case 5. Likewise, energy turbined drops by 27.8 %, from 656,319 MWh to 473,622 MWh. Consequently, as the model becomes more flexible, it prioritizes the purchase of electricity at low-cost hours on the day-ahead electricity market rather than storing surplus renewable energy. It should be taken into account that there is less pressure to comply with the demand coverage restriction with renewables; therefore, the use of the electrical grid can be taken advantage of when it is more profitable than the use of other available resources.

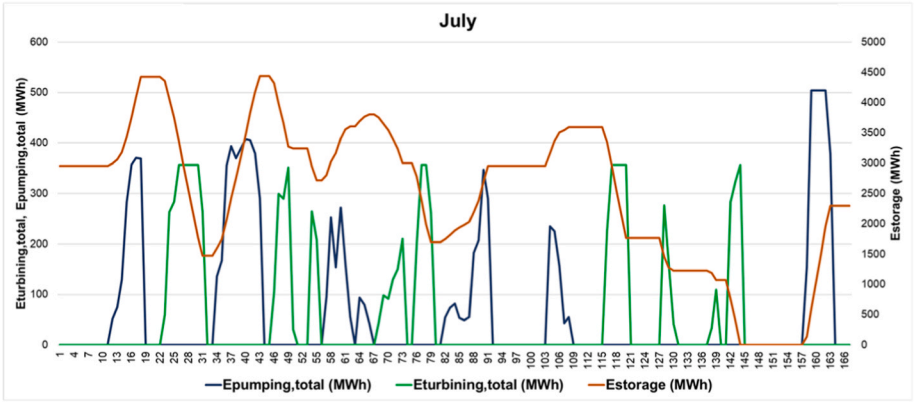
In Case 5, where the allowed risk is higher, the model seeks the optimal management of the available resources (production, storage and grid connection) to obtain the maximum profit. Therefore, in low-price hours of the electricity market or excess generation, the system stores energy, using electricity to pump water to the upper reservoir. On the



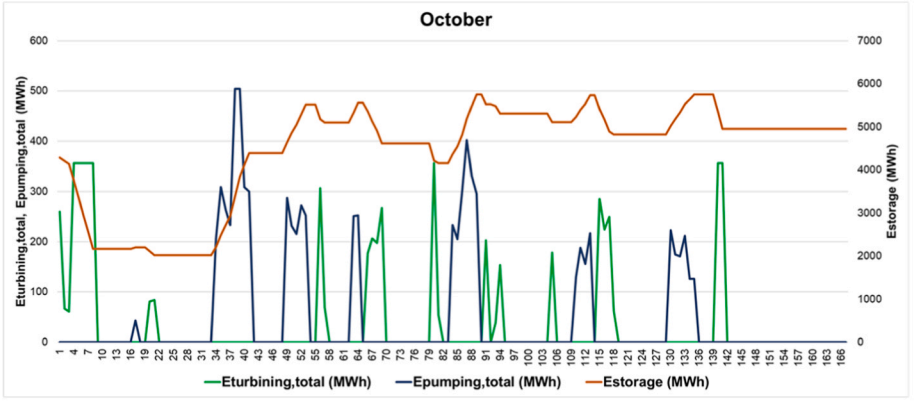
a) Pumped hydro storage management for one week in January



b) Pumped hydro storage management for one week in April

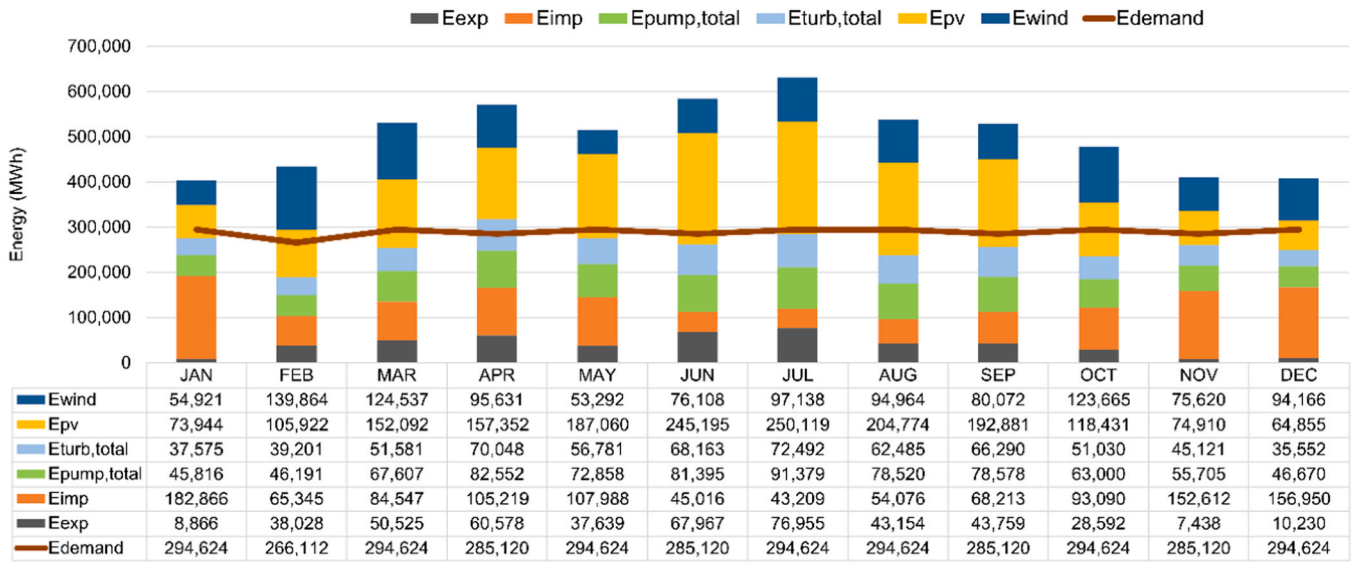


c) Pumped hydro storage management for one week in July



d) Pumped hydro storage management for one week in October

Fig. 6. Optimal hourly pumped hydro storage results for one week of different months (Case 1).



	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Ewind	54,921	139,864	124,537	95,631	53,292	76,108	97,138	94,964	80,072	123,665	75,620	94,166
Epv	73,944	105,922	152,092	157,352	187,060	245,195	250,119	204,774	192,881	118,431	74,910	64,855
Eturb,total	37,575	39,201	51,581	70,048	56,781	68,163	72,492	62,485	66,290	51,030	45,121	35,552
Epump,total	45,816	46,191	67,607	82,552	72,858	81,395	91,379	78,520	78,578	63,000	55,705	46,670
Eimp	182,866	65,345	84,547	105,219	107,988	45,016	43,209	54,076	68,213	93,090	152,612	156,950
Eexp	8,866	38,028	50,525	60,578	37,639	67,967	76,955	43,154	43,759	28,592	7,438	10,230
Edemand	294,624	266,112	294,624	285,120	294,624	285,120	294,624	294,624	285,120	294,624	285,120	294,624

Fig. 7. Monthly energy results (Case 1).

Table 3
Annual energy results obtained (MWh).

Case	1	2	3	4	5
E_{exp}	473,731	395,173	338,133	302,044	273,203
E_{imp}	1,158,736	1,243,995	1,304,866	1,363,798	1,413,979
E_{PV}	1,827,533	1,715,556	1,617,581	1,540,805	1,472,737
E_{wind}	1,109,977	1,044,767	1,013,318	986,661	966,148
$E_{pumping,total}$	810,271	739,899	679,304	635,029	584,718
$E_{turbining,\ total}$	656,319	599,318	550,237	514,373	473,622
E_{demand}	3,468,960	3,468,960	3,468,960	3,468,960	3,468,960
% demand coverage with renewables	90	86	82	79	75

other hand, in hours with high market prices or insufficient generation to meet demand, the system releases the stored water (through the turbining process) to generate electricity, which is injected into the grid and additional income is obtained.

On the contrary, Case 1 where there is less risk allowed (α=5 % and β=10 %), the system is more autonomous; therefore, the use of renewable generation and storage plants is optimized to meet the imposed demand coverage with renewable production.

In short, it should be noted that as the allowed risk increases, the probability of meeting demand coverage with renewables decreases. This reflects the increased tolerance for non-compliance with renewable coverage, which allows the system to prioritize economic decisions over sustainability objectives.

5.3. Annual economic results

Table 4 shows the optimal annual economic results obtained for the different cases proposed. As can be observed, as the analysis progresses from Case 1 to Case 5, the system's total costs progressively decrease. This reduction is largely due to greater flexibility in managing available resources, which allows for optimizing technical and economic operation of the system. In particular, in Case 5, the cost reduction is due to the ability to take advantage of hours with more competitive market prices to meet demand, replacing renewable generation during periods of lower profitability. Generation costs are reduced as a result of the decrease in energy produced as a result of its low profitability at certain times (a reduction of 28 % for both technologies compared to Case 1). Therefore, in the latter case, economic optimal management strategies are prioritized, as opposed to Case 1 where demand coverage with renewable production must be met with a 95 % probability. Regarding import costs, they are decreased by 40 %. This trend reflects that, even though low-price hours are taken advantage of, it is still necessary to purchase energy in the electricity market to meet demand at all times.

On the other hand, although the absolute costs of pumping and turbine generation decrease, their percentage in relation to import costs increases. For example, in Case 1, pumping represents 11.31 % of the import cost, while in Case 5 it reaches 18.79 %. Similarly, turbine generation increases from 12.74 % to 23.11 %. This has an important significance because it shows that, as import dependence decreases, pumped hydro storage assumes a more relevant role in meeting demand during off-peak renewable production hours.

The comparison between the different cases reflects the necessary balance between economic efficiency and system reliability. While Case 1 prioritizes demand coverage with renewables, which implies higher costs, Case 5 opts for a more flexible and economical management, although with a higher risk of not meeting demand with a certain percentage of renewables in its entirety.

6. Conclusions

The incorporation of wind uncertainty management is key for

Table 4
Annual economic results obtained (euros).

Case	C_{exp}	C_{imp}	C_{PV}	C_{wind}	$C_{pumping,total}$	$C_{turbining,\ total}$
1	53,059,556	104,071,387	2,835,948	2,297,118	14,620,265	16,649,658
2	29,202,925	74,713,025	2,589,645	2,097,613	13,724,451	15,671,505
3	24,842,152	69,496,899	2,377,566	1,925,828	12,940,649	15,199,769
4	22,284,351	66,539,758	2,222,601	1,800,307	12,326,440	14,799,910
5	19,825,626	62,688,964	2,046,513	1,657,676	11,781,892	14,492,215

effective local scheduling and operation of renewable energy assets, due to their variable and intermittent nature. In this context, this article develops an optimal hourly mathematical model that aims to maximize the monthly operating profit of an energy system that includes wind and photovoltaic generation, reversible pumped hydro storage, and grid connection. In addition, the uncertainty management of the wind resource is incorporated, using the chance-constrained method, to analyze the level of risk in the optimal management of energy assets in the system and guarantee a certain percentage of demand coverage is reliably met with renewable sources and storage. The optimization is extended throughout the twelve months of the year to provide a comprehensive and realistic assessment of the system's operational strategy.

The results show a strong seasonality in the system's operations, mainly marked by photovoltaic energy. In months of higher generation, the system minimizes the purchase of energy from the grid, while maximizing the use of storage and export. Conversely, in months of low renewable production, the system relies more on the grid to meet the required demand.

The integration of renewable energy and reversible pumped hydro storage allows for efficient system operation. In the best-case scenario, with 95 % reliability, wind generation accounts for approximately 65 % of total renewable energy production, while photovoltaic contributes 15 %. However, as the level of risk increases and reliability decreases to 75 %, wind generation drops to 55 %, and PV decreases to 10 %, increasing the need to import power from the grid to meet demand.

In addition, it has been demonstrated that in lower risk scenarios, the model prioritizes meeting local demand with renewables, which implies maximizing electricity generation from photovoltaic and wind power plants. As the allowed risk increases, the flexibility of the model allows for greater reliance on imported power rather than local production at times when operating renewables has a higher cost.

Although the amount of imported energy rises as risk increases (18 % increment), the system takes advantage of the lower prices in the electricity market to purchase energy in these hours and maximize the economic profit. In this way, energy import costs are reduced by 40 %. Therefore, the flexibility of the system makes it possible to optimize all available resources at minimum cost.

In short, the proposed model constitutes an essential tool for the efficient management of energy systems with the integration of renewables and storage, as it allows incorporating the uncertainty inherent to the wind resource in operational decisions. In this way, the model provides an adequate basis for effective energy planning. This approach not only maximizes economic profit, but also guarantees a certain demand coverage with renewable generation, enhancing the system's overall sustainability.

CRediT authorship contribution statement

Natalia Naval: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Jose M. Yusta:** Validation, Supervision, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] IHA, "Hydropower Status Report," 2022.
[2] M.S. Javed, T. Ma, J. Jurasz, M.Y. Amin, Solar and wind power generation systems with pumped hydro storage: review and future perspectives, Renew. Energy 148 (2020) 176–192, https://doi.org/10.1016/j.renene.2019.11.157.
[3] A. Emrani, A. Berrada, A comprehensive review on techno-economic assessment of hybrid energy storage systems integrated with renewable energy, J. Energy Storage 84 (PB) (2024) 111010, https://doi.org/10.1016/j.est.2024.111010.
[4] M. Petrollese, P. Seche, D. Cocco, Analysis and optimization of solar-pumped hydro storage systems integrated in water supply networks, Energy 189 (2019) 116176, https://doi.org/10.1016/j.energy.2019.116176.
[5] S. Lin, T. Ma, M. Shahzad Javed, Prefeasibility study of a distributed photovoltaic system with pumped hydro storage for residential buildings, Energy Convers. Manag. 222 (February) (2020) 113199, https://doi.org/10.1016/j.enconman.2020.113199.
[6] P. Cruz, H.M.I. Pousinho, R. Melício, V.M.F. Mendes, Optimal coordination on wind-pumped-hydro operation, Procedia Technol. 17 (2014) 445–451, https://doi.org/10.1016/j.protcy.2014.10.253.
[7] S.V. Papaefthymiou, S.A. Papathanassiou, Optimum sizing of wind-pumped-storage hybrid power stations inisland systems, Renew. Energy 64 (2014) 187–196, https://doi.org/10.1016/j.renene.2013.10.047.
[8] E.N. Nyeche, E.O. Diemuodeke, Modelling and optimisation of a hybrid PV-wind turbine-pumped hydro storage energy system for mini-grid application in coastline communities, J. Clean. Prod. 250 (2020) 119578, https://doi.org/10.1016/j.jclepro.2019.119578.
[9] S. Makhdoomi, A. Askarzadeh, Optimizing operation of a photovoltaic/diesel generator hybrid energy system with pumped hydro storage by a modified crow search algorithm, J. Energy Storage 27 (September) (2020) 101040, https://doi.org/10.1016/j.est.2019.101040.
[10] B.A. Bhayo, H.H. Al-Kayiem, S.I.U. Gilani, F.B. Ismail, Power management optimization of hybrid solar photovoltaic-battery integrated with pumped-hydro-storage system for standalone electricity generation, Energy Convers. Manag. 215 (May) (2020) 112942, https://doi.org/10.1016/j.enconman.2020.112942.
[11] X. Liu, N. Li, H. Mu, M. Li, X. Liu, Techno-energy-economic assessment of a high capacity offshore wind-pumped-storage hybrid power system for regional power system, J. Energy Storage 41 (December) (2021) 102892, https://doi.org/10.1016/j.est.2021.102892.
[12] S. Makhdoomi, A. Askarzadeh, Daily performance optimization of a grid-connected hybrid system composed of photovoltaic and pumped hydro storage (PV/PHS), Renew. Energy 159 (2020) 272–285, https://doi.org/10.1016/j.renene.2020.06.020.
[13] U. Yıldıran, İ. Kayahan, Risk-averse stochastic model predictive control-based real-time operation method for a wind energy generation system supported by a pumped hydro storage unit, Appl. Energy 226 (October) (2018) 631–643, https://doi.org/10.1016/j.apenergy.2018.05.130.
[14] P.T. Kyrimlidou, C.K. Simoglou, P.N. Biskas, Optimal sizing and long-term operation of a hybrid RES-BESS station in a competitive electricity market, Sustain. Energy Grids Netw. 39 (June) (2024) 101450, https://doi.org/10.1016/j.segan.2024.101450.
[15] Y. Wang, Y. Zheng, Q. Yang, Optimal energy management of integrated energy systems for strategic participation in competitive electricity markets, Energy 278 (PA) (2023) 127883, https://doi.org/10.1016/j.energy.2023.127883.
[16] L. Hou, et al., Optimized scheduling of smart community energy systems considering demand response and shared energy storage, Energy 295 (December) (2024) 131066, https://doi.org/10.1016/j.energy.2024.131066.
[17] R.J. Mahfoud, N.F. Alkayem, Y. Zhang, Y. Zheng, Y. Sun, H.H. Alhelou, Optimal operation of pumped hydro storage-based energy systems: a compendium of current challenges and future perspectives, Renew. Sustain. Energy Rev. 178 (March) (2023) 113267, https://doi.org/10.1016/j.rser.2023.113267.
[18] Y. Li, F. Wu, J. Li, Y. Yin, Z. Li, L. Ai"Chance-constrained energy management for pumped storage hydropower plant to compensate for wind power uncertainties," In: Proceedings of the IEEE Power and Energy Society General Meeting, vol. 2021-July, pp. 1–5, 2021, doi: 10.1109/PESGM46819.2021.9637867.
[19] U. Portero, S. Velázquez, J.A. Carta, Sizing of a wind-hydro system using a reversible hydraulic facility with seawater. A case study in the canary islands, Energy Convers. Manag. 106 (2015) 1251–1263, https://doi.org/10.1016/j.enconman.2015.10.054.
[20] A. Golshani, W. Sun, Q. Zhou, Q.P. Zheng, J. Wang, F. Qiu, Coordination of wind farm and pumped-storage hydro for a self-healing power grid, IEEE Trans. Sustain. Energy 9 (4) (2018) 1910–1920, https://doi.org/10.1109/TSTE.2018.2819133.
[21] U. Yıldıran, İ. Kayahan, Risk-averse stochastic model predictive control-based real-time operation method for a wind energy generation system supported by a pumped hydro storage unit, Appl. Energy 226 (May) (2018) 631–643, https://doi.org/10.1016/j.apenergy.2018.05.130.
[22] A.V. Ntomaris, A.G. Bakirtzis, Stochastic scheduling of hybrid power stations in insular power systems with high wind penetration, IEEE Trans. Power Syst. 31 (5) (2016) 3424–3436, https://doi.org/10.1109/TPWRS.2015.2499039.
[23] P. Toufani, E. Nadar, A.S. Kocaman, Short-term assessment of pumped hydro energy storage configurations: up, down, or closed? Renew. Energy 201 (P1) (2022) 1086–1095, https://doi.org/10.1016/j.renene.2022.11.009.
[24] A. Ghasemi, M. Enayatzare, Optimal energy management of a renewable-based isolated microgrid with pumped-storage unit and demand response, Renew. Energy 123 (2018) 460–474, https://doi.org/10.1016/j.renene.2018.02.072.
[25] D. Fernández-Muñoz, J.I. Pérez-Díaz, M. Chazarra, A two-stage stochastic optimisation model for the water value calculation in a hybrid diesel/ wind/ pumped-storage power system, IET Renew. Power Gener. 13 (12) (2019) 2156–2165, https://doi.org/10.1049/iet-rpg.2018.6151.

[26] J. Jurasz, Modeling and forecasting energy flow between national power grid and a solar–wind–pumped-hydroelectricity (PV–WT–PSH) energy source, Energy Convers. Manag. 136 (2017) 382–394, https://doi.org/10.1016/j.enconman.2017.01.032.
[27] A.S. Kocaman, V. Modi, Value of pumped hydro storage in a hybrid energy generation and allocation system, Appl. Energy 205 (April) (2017) 1202–1215, https://doi.org/10.1016/j.apenergy.2017.08.129.
[28] A.J. Conejo, M. Carrión, and J.M. Morales, "Decision making under uncertainty in electricity markets," vol. 153, 2010, doi: 10.1007/978-1-4419-7421-1.
[29] P. Toufani, E.C. Karakoyun, E. Nadar, O.B. Fosso, A.S. Kocaman, Optimization of pumped hydro energy storage systems under uncertainty: a review, J. Energy Storage 73 (PD) (2023) 109306, https://doi.org/10.1016/j.est.2023.109306.
[30] R. Lu, T. Ding, B. Qin, J. Ma, X. Fang, Z. Dong, Multi-stage stochastic programming to joint economic dispatch for energy and reserve with uncertain renewable energy, IEEE Trans. Sustain. Energy 11 (3) (2020) 1140–1151, https://doi.org/10.1109/TSTE.2019.2918269.
[31] P. Aaslid, M. Korpas, M.M. Belsnes, O.B. Fosso, Stochastic optimization of microgrid operation with renewable generation and energy storages, IEEE Trans. Sustain. Energy 13 (3) (2022) 1481–1491, https://doi.org/10.1109/TSTE.2022.3156069.
[32] X. Lu, Z. Liu, L. Ma, L. Wang, K. Zhou, N. Feng, A robust optimization approach for optimal load dispatch of community energy hub, Appl. Energy 259 (November) (2020), https://doi.org/10.1016/j.apenergy.2019.114195.
[33] J. Yang, C. Su, Robust optimization of microgrid based on renewable distributed power generation and load demand uncertainty, Energy 223 (2021) 120043, https://doi.org/10.1016/j.energy.2021.120043.
[34] E.D. Castronuovo, et al., An integrated approach for optimal coordination of wind power and hydro pumping storage, Wind Energy 17 (6) (2014) 829–852, https://doi.org/10.1002/we.1600.
[35] N. Lu, C. Su, C. Guo, P. Wang, H. Wu, Q. Sui, Stochastic optimal scheduling of wind power and pumped-storage hydropower complementary systems with multiple uncertainties, J. Energy Storage 78 (June 2023) (2024), https://doi.org/10.1016/j.est.2023.110060.
[36] H. Ding, Z. Hu, Y. Song, Rolling optimization of wind farm and energy storage system in electricity markets, IEEE Trans. Power Syst. 30 (5) (2015) 2676–2684, https://doi.org/10.1109/TPWRS.2014.2364272.
[37] Y.Y. Hong, G.F.D. Apolinario, T.K. Lu, C.C. Chu, Chance-constrained unit commitment with energy storage systems in electric power systems, Energy Rep. 8 (2022) 1067–1090, https://doi.org/10.1016/j.egyr.2021.12.035.
[38] J. Cheng, R.L.Y. Chen, H.N. Najm, A. Pinar, C. Safta, J.P. Watson, Chance-constrained economic dispatch with renewable energy and storage, Comput. Optim. Appl. 70 (2) (2018) 479–502, https://doi.org/10.1007/s10589-018-0006-2.
[39] A. Zhou, M. Yang, Z. Wang, P. Li, A linear solution method of generalized robust chance constrained real-time dispatch, IEEE Trans. Power Syst. 33 (6) (2018) 7313–7316, https://doi.org/10.1109/TPWRS.2018.2865184.
[40] International Hydropower Association, "Innovative pumped storage hydropower configurations and uses," no. September, 2021.
[41] W. Yang, J. Yang, Advantage of variable-speed pumped storage plants for mitigating wind power variations: integrated modelling and performance assessment, Appl. Energy 237 (September 2018) (2019) 720–732, https://doi.org/10.1016/j.apenergy.2018.12.090.
[42] I. Sârbu, I. Borza, Energetic optimization of water pumping in distribution systems, Period. Polytech. Ser. Mech. Eng. 42 (4) (1998) 141–152.
[43] A.R. Jordehi, How to deal with uncertainties in electric power systems? A review, Renew. Sustain. Energy Rev. 96 (August) (2018) 145–155, https://doi.org/10.1016/j.rser.2018.07.056.
[44] H. Ding, Z. Hu, Y. Song, Stochastic optimization of the daily operation of wind farm and pumped-hydro-storage plant, Renew. Energy 48 (2012) 571–578, https://doi.org/10.1016/j.renene.2012.06.008.
[45] J.F. Toubeau, et al., Non-linear hybrid approach for the scheduling of merchant underground pumped hydro energy storage, IET Gener. Transm. Distrib. 13 (21) (2019) 4798–4808, https://doi.org/10.1049/iet-gtd.2019.0204.
[46] "Law 15/2012. Tax on the value of electricity production." [Online]. Available: ⟨https://www.boe.es/buscar/doc.php?id=BOE-A-2012-15649⟩. [Accessed: 07-Apr-2024].
[47] H. ZechunEnergy storage for power system planning and operation. 2020..
[48] F. Marzbani, A. Abdelfatah, Economic dispatch optimization strategies and problem formulation: a comprehensive review, Energies (2024) 1–31.
[49] J.F. Toubeau, F. Vallée, K. Bruninx, Chance-Constrained scheduling of underground pumped hydro energy storage in presence of model uncertainties, IEEE Trans. Sustain. Energy 11 (3) (2020) 1516–1527.